

Nuclear equations

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	Describe how the mass number and atomic number change when an alpha particle is emitted.	Mass number decreases by 4. Atomic number decreases by 2.	☐
2 ★	Describe how the mass number and atomic number change when a beta particle is emitted.	Mass number stays the same. Atomic number increases by 1.	☐
3	Explain why gamma emission does not change the mass number or atomic number.	A gamma ray has no mass and no charge.	☐
4	State the symbol for an alpha particle in a nuclear equation.	${}^4_2\text{He}$	☐
5	State the symbol for a beta particle in a nuclear equation.	${}^0_{-1}\text{e}$	☐
6 ★	State the two quantities that must balance in a nuclear equation.	The total mass number and the total atomic number	☐
7	State what happens to the element when a nucleus emits an alpha or beta particle.	It becomes a different element, because the number of protons changes.	☐